NANYANG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE NAME

CLASS

BIOLOGY 9648/03

Paper 3 Applications and Planning **September 2011**

2 hours

Candidates answer questions 1 to 4 on the Question Paper.
Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your name and CT on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use soft pencil for any diagrams, graphs or rough working.
Do no use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

Answer questions **1** to **4** in the spaces provided on the question paper.

Answer question **5** on the separate answer paper provided.

For Examiner's Use	
1	
2	
3	
5	
Total	60
For Examiner's Use	
4	12

This document consists of **14** printed pages and **1** blank page.

[Turn over

For examiner's use

Section A

Answer **all** the questions in this section.

1 **(a)** Adenosine deaminase (ADA)-deficient SCID is an autosomal recessive genetic disease which affects the immune system in a detrimental manner.

(i) Describe what "autosomal recessive" inheritance of ADA-deficient SCID means.

Sufferers must be homozygotes/ possess two recessive alleles/ homozgyous;

Both parents have to be carriers;

offspring has 1 in 4 chance of getting the disease;

[2]

(ii) Explain how ADA deficiency results in a SCID.

Mutated adenosine deaminase cannot break down purines;

Accumulation of purines/ toxic metabolites inhibits enzymes involved in lymphocyte proliferation, lymphocyte cells decrease;

antibodies are not produced against foreign pathogens / pathogens not recognised and killed / increase in opportunistic infections;

[3]

(b) Scientists are investigating the use of stem cells in monkeys to treat a CFTR mutation which causes cystic fibrosis in monkeys.

(i) Explain how a mutation in the CFTR gene leads to lung problems in cystic fibrosis sufferers.

deletion of 3 bases from CFTR gene deletes phenylalanine in protein;

changes the 3D conformation of channel protein;

mutated protein does not transport chloride ions out of the cell, increasing the solute concentration of cell/ water potential becomes more negative;

water moves from mucus on the extracellular surface into the cell via osmosis and thick mucus forms;

thick mucus interferes with gaseous exchange/diffusion / tends to trap bacteria leading to increased infections;

[3]

For examiner's use

An experiment utilizing genetic engineering techniques was conducted to obtain genetically modified stem cells from monkeys suffering from cystic fibrosis.

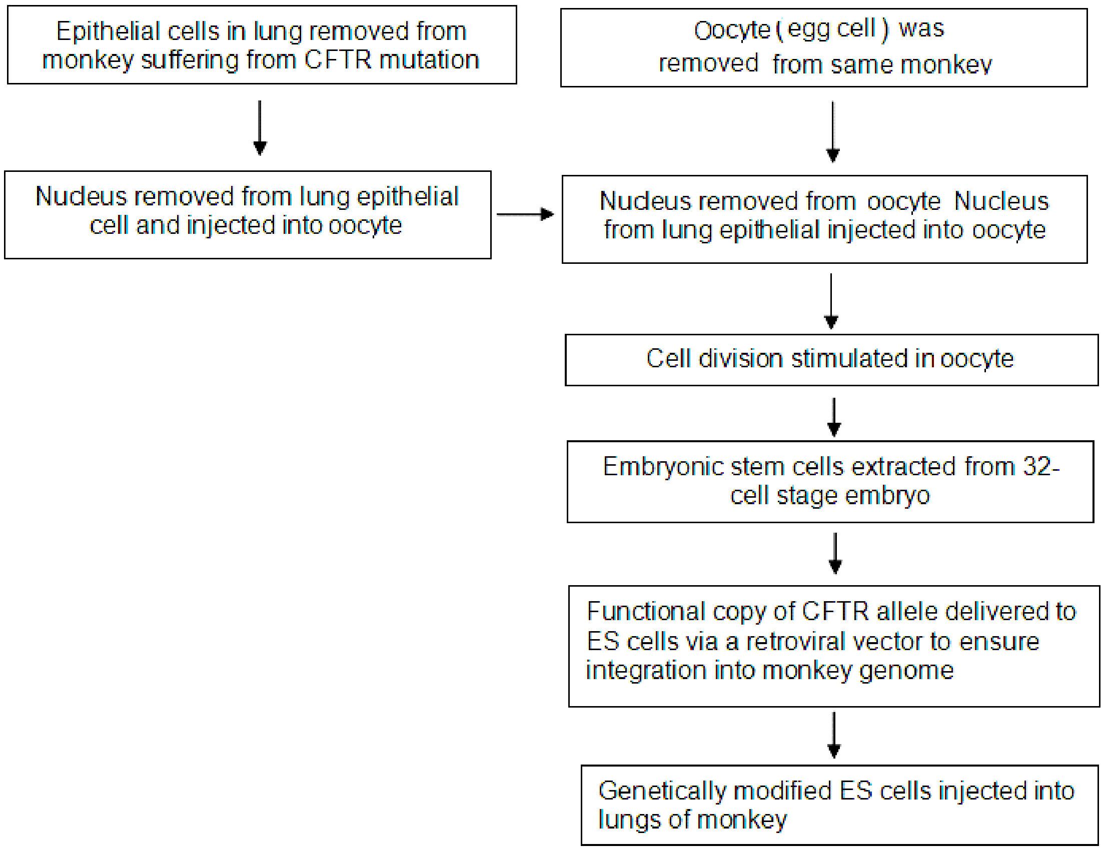


Fig. 1.1

With reference to **Fig. 1.1**,

(ii) describe the characteristics of the ES cells produced.

Self-renewal property: able to divide into 2 cells by mitosis, with one remain unspecialized while the other differentiate;

Pluripotent : stem cells that can differentiate into almost any type of cell in the body;

Can differentiate/specialize due to differential gene expression/ different sets of genes activated;

[2]

(iii) suggest why an oocyte was used and why the nucleus had to be removed from the oocyte.

oocyte is the only cell with potential to develop into a ball of cells/ blastocyst in which all the cells are pluripotent;

haploid oocyte nucleus is removed so a diploid nucleus can be added/ transferred;

Oocyte contains the organelles necessary for cell division and growth;

[2]

[Total: 12]

2 A disease-causing allele is closely linked to an RFLP as shown in **Fig. 2.1**. Two RFLPs are identified for the same loci. **Fig. 2.2** shows a pedigree of a family for the disease and the linkage map for the RFLP locus on a Southern blot.

Fig. 2.1

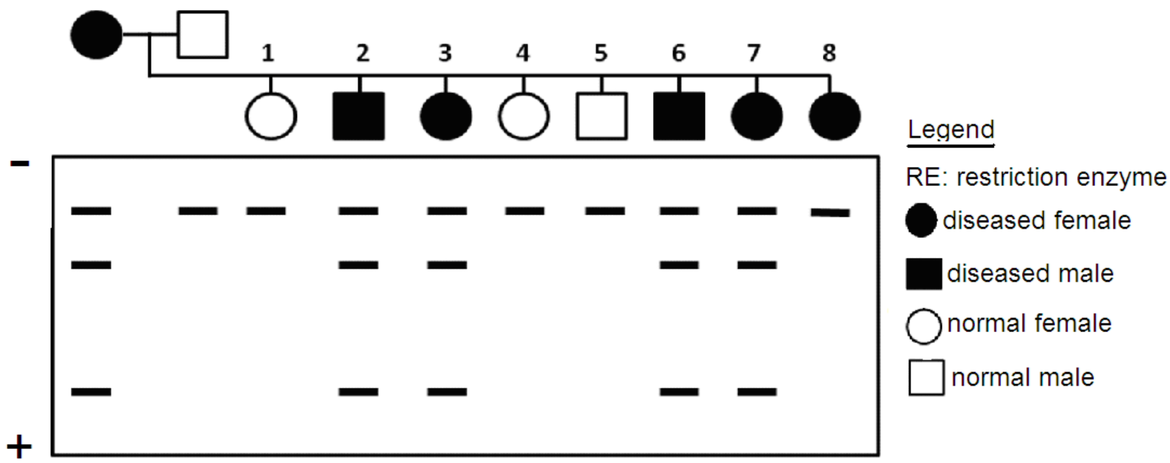
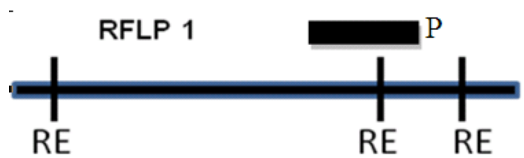


Fig. 2.2

(a) The 2 RFLPs are detected using a probe **P**. On RFLP **1** of **Fig. 2.1**, indicate the position that probe **P** will bind.



[1]

(b) With reference to **Fig 2.1** and **Fig 2.2,**

(i) explain which RFLP the disease-causing allele linked is to;

RFLP 1;

All affected individuals with exception of 8 have the 2 shorter fragments produced from cutting RFLP 1;

[2]

(ii) explain if the disease is caused by a dominant or recessive allele;

Dominant ;

Mother /individual 2/3/6/7, are heterozygous for the RFLP, possesses both RFLP 1 & 2, is diseased;

Father is homozygous for RFLP 2 and is disease-free;

only 1 copy of allele linked to RFLP 1 is required to mask the expression of allele linked to RFLP 2;

[3]

(iii) explain how the banding pattern observed in individual 8 arose.

Crossing over occurred at a locus between the allele and RFLP marker;

Dominant allele is no longer linked to RFLP 1;

[2]

One of the applications of RFLP analysis is for restriction mapping. *Bam*HI, *Eco*RI and *Hind*III restriction enzymes are used to map a 15kb bacterial DNA. Three reactions were set up to cut the 15kb DNA sample, each with one of the three enzymes. After digestion with restriction enzymes, the DNA fragments were separated into bands as shown in **Fig. 2.3**.

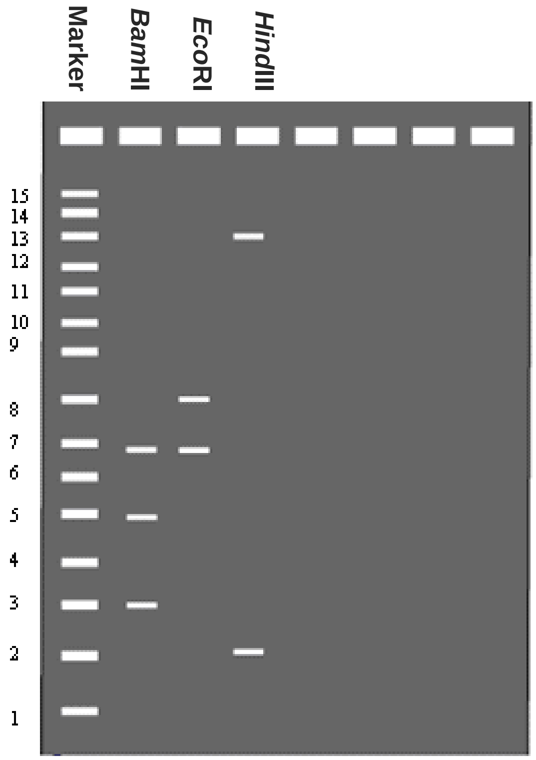


Fig. 2.3

(c) Describe briefly how, in such DNA analysis,

(i) DNA is separated into bands.

via gel electrophoresis;

negatively charged DNA migrate towards positive electrode / anode;

DNA fragments thus separated by their molecular size / length ;

smaller / shorter restriction fragments migrate faster / move a longer distance <u>in a fixed time</u>;

[2]

(ii) the bands are made visible.

stained with <u>ethidium bromide</u> which intercalate between DNA and glows <u>under UV light;</u>

OR Methylene blue

[1]

For examiner's use

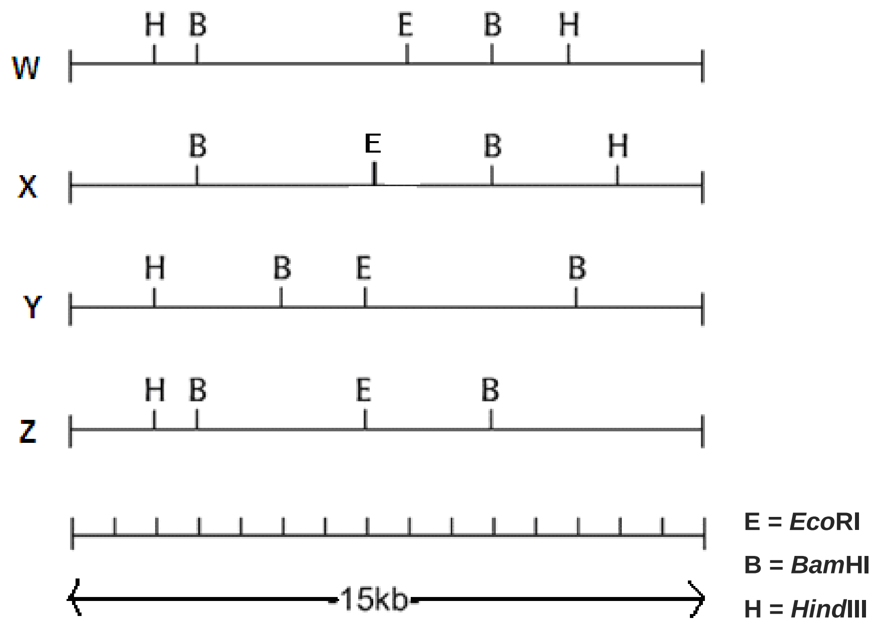


Fig. 2.4

(iii) Based on the results shown in **Fig. 2.3** and **Fig. 2.4**, explain which of the following map (**W**, **X**, **Y** or **Z**) in **Fig. 2.4** is **NOT** a possible map.

W;

HindIII cuts at 2 sites, giving rise to 3 fragments, however, only 2 fragments for HindIII are detected in Fig. 2.1.

[2]
[Total: 13]

3 The detrimental effect of high salt levels to plants usually cause death or severely stunted growth. A gene encoding a vacuolar Na^+ channel protein from a weed, *Arabidopsis thaliana* was cloned into tomatoes *Lycopersicon esculentum*. Transgenic tomato plants were able to flower and fruit when grown in high salinity conditions.

(a) Suggest the natural role of the vacuolar Na^+ channel protein.

found on the tonoplast/ vacuole membrane;

Transports Na^+ from cytoplasm into vacuole of plant cell OR stores / sequesters Na+ in vacuole;

maintains water potential/ solute potential of cell/ cytoplasm;

[2]

(b) An experiment was conducted to study the ability to survive on high salinity soil. Twenty plants were used for each condition at the beginning of the experiment. Results of the experiment were shown in **Table 3.1** below.

Type of plant	**Number of plants that survived after transplanting to soil containing NaCl**	
	5 mmol dm^{-3} NaCl	200 mmol dm^{-3} NaCl
Wild type	20	1
Transgenic	20	19

Table 3.1

With reference to **Table 3.1**, describe the survival rate of wild type and transgenic plants in soil containing NaCl

Both wild type and transgenic plants, all plants survive at 5 mmol dm^{-3} NaCl / 100% survival;

1 out of 20 of wild type plants in soil containing 200 mmol dm^{-3} NaCl survived / 5% survival rate;

19 out of 20 transgenic plants survived in soil containing 200 mmol dm^{-3} NaCl / 95% survival rate;

[3]

For examiner's use

The transgenic plants were grown from tissue culture after successful transformation using a T-plasmid containing a constitutive leaf-specific promoter. Experiments were done to extract the contents of vacuoles of both the wild type and transgenic plants for analysis. The following data was presented.

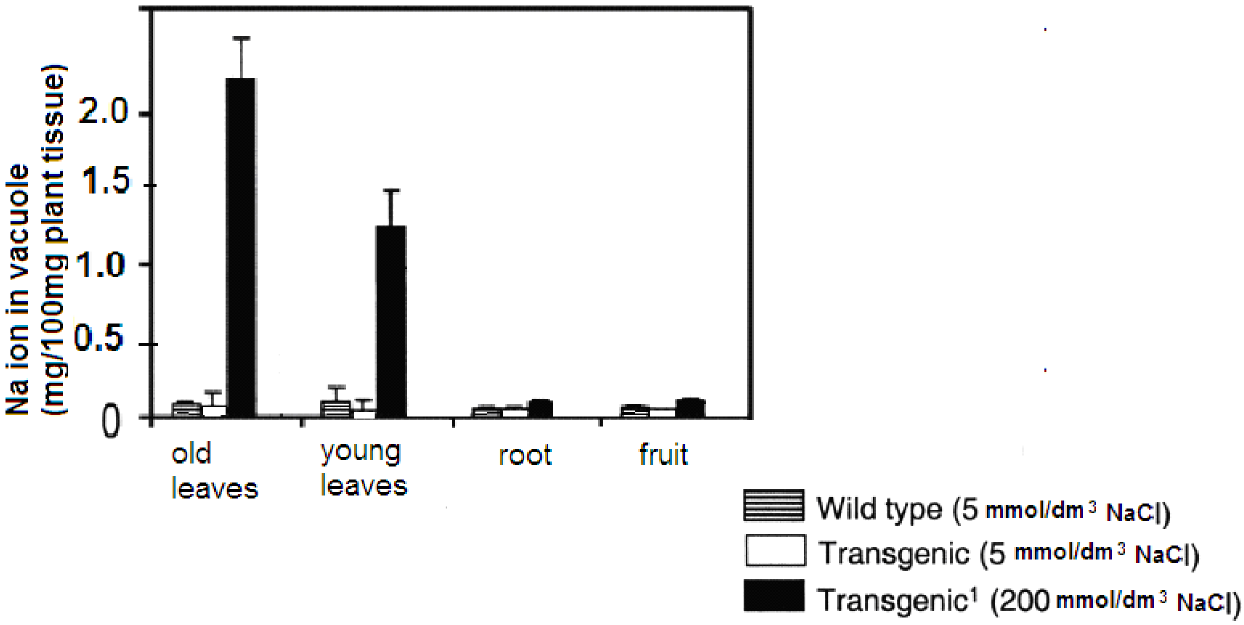


Fig. 3.2

(c) With reference to **Fig. 3.2**,

(i) describe the data observed in leaves;

Both wild type and transgenic plants, young or old leaves grown at 5 mmol dm^{-3} NaCl had very low levels of 0.1mg Na /100 mg plant tissue ;

For transgenic plants grown 200 mmol dm^{-3} NaCl, old leaves had 2.2 mg Na while young leaves had 1.25 mg Na;

[2]

(ii) explain the data observed in leaves;

Transgenic plants had more sodium channels embedded in tonoplast/ membrane allowing salt to be sequestered;

not much Na transported into vacuole for transgenic plants grown at 5 mmol dm^{-3} NaCl / NaCl conc is low in environment;

Transgenic old leaves formed earlier and had accumulated more NaCl over time;

[2]

(iii) suggest the effectiveness of the constitutive leaf-specific promoter in the expression of the transgene;

leaf specific promoter allows sodium channel to be specifically transcribed and translated in leaves and not other parts of the plants;

highly expressed all the time and embedded in the vacuole membrane;

For examiner's use

For transgenic plants grown 200 mmol dm^{-3} NaCl, the vacuole content from root and fruit cells remains low at 0.2 mg, indicating the promoter is working/ functional;

root and fruit cells' tonoplasts lack sodium channel and sodium ions not accumulated;

[3]

(iv) suggest why there were no comparison of wild type plants grown in soil with 200 mmoldm^{-3};

Wild type plants grown 200 mmol dm^{-3} NaCl died or did not develop properly;

[1]

(d) Suggest **two** benefits of this genetically engineered tomato plant.

possible to use land which has high salt content which previously cannot sustain agricultural crops OR Use plant to convert non-arable high salinity ground into arable land;

Increase food yield to solve food shortage issue;

harvest tomato fruits which are not salty as Na is not sequestered into vacuoles of fruit cells;

Grown together with low salt tolerant plants to absorb salt content from ground to increase food yield;

[2]

[Total: 15]

For examiner's use

4 Skill A – Planning

You are required to plan, but not carry out, an investigation on the difference in duration of each stage in the mitotic cycle, with respect to one another, in an actively dividing cell. This is to establish the longest stage in mitosis.

The mitotic activity of a root tip cell can be estimated by observing cells undergoing cell division with a microscope. When examined under a microscope, the percentage of cells that are undergoing mitosis at different stages can be recorded.

Your planning must be based on the assumption that you have been provided with the following equipment and materials which you should use:

- a microscope
- onion roots with intact tips
- toluidine blue which stains chromosomes blue
- hydrochloric acid at 60°C which fixes the specimen

Your plan should have a clear and helpful structure to include:

- an explanation of the theory to support your practical procedure
- a description of the method used, including the scientific reasoning behind the method and any recommended safety measures
- an explanation of the dependant and independent variables involved
- relevant, clearly labelled diagrams
- how you will record your results and ensure that they are as accurate and reliable as possible
- proposed layout of results tables and graphs with clear headings and labels
- the correct use of technical and scientific terms

[Total: 12]

For examiner's use

Mark Scheme for Marking

1. Theoretical basis (root tip actively growing, must choose zone of elongation)
2. Theoretical basis (describe mitosis)
3. Independent variables : stage of mitosis
   dependent variable : number of cells, which is used to calculate time taken to complete stage;
   Constant variable : use of same root tip / use of onions of similar stages /different root tips at similar development stage
4. Hypothesis : more cells at a stage, longer time taken to complete stage/ longest stage
5. Description of sample / slide preparation. handle the root tip by its cut end to reduce damage to dividing cells, make a flatten layer of cell, stain makes chromosomes visible
6. HCl kills cells, hence fixing chromosomes in its stage for observation
7. Description of use of microscope; start with lowest magnification, 40x/ scanning power and increase magnification to at least x400/ high power)
8. Description/ Labelled diagram of identifying the different stages
9. Description of how percentage of cells in different stage of mitosis will be recorded;
   (give formula/ describe)
10. Repeats (FOV) to increase reliability x2 / 3 readings
11. Table of results with correct headers
12. Processing of data; (1) calculate % of each stage (2) calculate mean % of each stage
13. Plot bar graph with correct labels
14. Conclusion: The <u>highest %</u> indicates the <u>longest stage</u> of mitosis which is <u>prophase</u>.
15. Description of safety precautions / risk assessment; reject glassware point
16. The correct use of technical and scientific terms;

For examiner's use

5 Write your answers to this question on the separate answer paper provided.
Your answer:

- should be illustrated by large, clearly labelled diagrams, where appropriate;
- must be in continuous prose, where appropriate;
- must be set out in sections **(a)**, **(b)**, etc., as indicated in the question.

(a) Distinguish between cDNA library and gDNA library. [6]

1 mark for each comparison (max 4)		
Property	**gDNA**	**cDNA**
Sequences	**Coding and non-coding**	**Coding**
	Uses **DNA in the nucleus**	Uses mature **mRNA**
Process	**Cut/ cleavage** by restriction enzymes to generate DNA fragments.	**Reverse transcription** mature mRNA to DNA called **complementary DNA/cDNA**.
Enzymes involved	**Restriction enzymes** and ligase	**Reverse transcriptase**, restriction enzymes and ligase
Fragment sizes	**Large/Long** 10-1000kbps	**Small/Short** 10-20kbps
Storage vectors	Phages, **bacterial artificial chromosome (BAC)** and **yeast artificial chromosome (YAC)**	**plasmids** and **phages**
Knowledge of source of gene of interest	Does not require prior knowledge of gene	Requires prior knowledge of gene sequence
One mark for each point from here onwards (max 2):		
General definition	Library contains **all sequences in genome** ;	Library contains a entire set of **genes** that are **being expressed** for for specialized function of a particular cell type. ;
Applications	**study the regulatory sequences or introns** of a gene ;	cDNA libraries of the same cell type at different times in the life of an organism can be used to **trace changes in patterns of gene expression** during development. ; **Identify specific genes** expressed by specific cell types ;

Max 6

For examiner's use

(b) Describe the aims and benefits of the human genome project. [8]

1. To obtain the **DNA sequence of the entire human genome** / determine the **sequences of the 3 billion chemical base pairs** that make up **human DNA**;

2. **identify all** the approximately 20,000-25,000 **genes in human DNA** by 2003;

3. **store** this **information in databases** and make it freely and totally accessible

4. To obtain a **physical map** of the human genome. This map shows the **exact distance between linked genes in a chromosome**.

5. To obtain a **genetic linkage map** of the human genome. This **map identifies the thousands of genetic markers** and their **location along the autosomes and sex chromosomes**.

6. **improve tools for data analysis**, especially for sequencing and comparison of genomes of other organisms;

7. **transfer related technologies to the private sector** and improve training and manpower;

8. **address the ethical, legal, and social issues** that may arise from the project;

Max 4

1. A detailed genetic and physical map has made it **easier for researchers to locate genes in order to study their functions**. The types of proteins encoded by these genes and their interaction in complex genetic diseases can be investigated.

2. HGP data can be used to **compare gene sequences between diseased and healthy individuals**. Hence mutations that contribute to disease can be identified and used as markers. The cloning and sequencing of disease-causing alleles is expected to play **important roles in the future diagnosis and treatment of disease**.

3. Sequences of **disease-causing alleles can be used in linkage studies**. By studying how inheritance of certain RFLPs correlate with specific cancer-causing alleles, HGP data can be used as a basis of detection of individuals possessing the diseased allele. Data can be used to advise individuals during genetic counseling.

4. HGP data can be used in **comparative studies with other model organisms** so that **disease development can be investigated using other model organisms**. Studying organisms *like Escherichia coli*, yeast, fruitfly and mouse helps researchers gain more insights about homologous genes, encoding proteins with similar functions in the human genome.

5. The **effect of non-coding DNA on gene expression can be investigated using sequences obtained from HGP**. E.g. mutation of control elements which cause diseases can be studied.

6. With the Human Genome Project, **technologies to manage computer data in user-friendly tools had been developed**. It allows researchers easy access to up-to-date information obtained from the project. Data processing, storage and analytical tools hare been developed to interpret and analyse sequence information in the field of bioinformatics and now are freely available

Max4

(c) Outline how plant tissue culture technique is used to produce transgenic plants. [6]

1. Plant tissue culture, a technique of **propagating plants under sterile conditions** is used;

2. Murashige and Skoog / growth **medium** (MS medium), containing sucrose, amino acids, inorganic ions and plant hormones / nutrients needed for plant growth is prepared. The medium is sterilized using high temperature. The liquid medium is cooled and solidified into an agar plate.
3. Antibiotics or herbicides are added for selection of successfully transformed cells..
4. **Plant hormones** are chemically substances produced by plants and they direct growth and differentiation of plants. In the medium, these hormones direct growth and differentiation of plant tissue: Auxin induces proliferation of roots, Cytokinin induces shoot formation. By adjusting the concentration of auxin and cytokinin in the growth medium, the cells in the callus can be induced to differentiate into roots and shoots.
5. An **explant is selected** for growth or culture. A small section of the shoot, root or leaf will usually be used because they contain meristematic cells which are able to divide and differentiate into any other type of cell.
6. The **transgene is inserted** into the explant cells via **electroporation / microprojectile bombardment / *Agrobacterium tumefaciens*-mediated transformation ;**
7. The explant is **surfaced sterilized** using alcohol or sodium hydrochlorite. The transfer chamber where the preparation is carried out and the apparatus such as forceps and scalpels also have to be sterilized. Transformed leaf or stem tissue is **placed on the medium agar**.
8. A **callus**, containing a mass of unorganized and undifferentiated cells is **formed by mitosis**. As the callus increase in size, **pieces of the callus is sliced off and grown on new medium composition in a process known as sub-culturing**.
9. **Plantlets** containing both roots and shoots are removed from agar medium and **transplanted to sterile soil for further growth**.